Astronomy Labs

SAMPLE

Alveary

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Alveary

Alt. Science Options

Astronomy Labs

GRADE(S): 11-12

About the Course

Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Astronomy

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
11-12	Astronomy Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Astronomy				
Astronomy Lessons Nature Walks & Scouting: Grades 9-12	Astronomy Lessons	Astronomy Lessons	Astronomy Lessons Nature Notebook: Grades 9-12	Astronomy Lessons Astronomy Labs

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Planning & Prep

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LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

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Quick Links

(No Quick Links Assigned)

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How To Approach Labs

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1 ⬚ **60m Astronomy Labs - Lesson 1**

Materials: journal, star finder

→ OBSERVE
Simply notice on this first night of observing. What are your first impressions? Or are you able to relocate any objects previously found? Do you notice anything new based on your use of the star finder or Quick Links? Keep a record of what you observe.

WEEK 2 ⬚ **60m Astronomy Labs - Lesson 2**

Materials: journal, star finder

→ OBSERVE
Notice any objects that are or are becoming familiar. In what directions do they currently lie? Using your hands, measure their locations in the sky. Do these locations change during the evening? Keep a record of what you observe.

★ STUDENT & TEACHER TIP
Note that next week's fieldwork should be done in one or more daytime sessions!

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 3**

Materials: journal, binoculars, paper, tripod/stand for binoculars

→ OBSERVE
Use your hands to measure locations and your binocular set-up to observe the sun's projection. DO NOT LOOK DIRECTLY AT THE SUN! Keep a record of what you observe.

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